

Model and data

Implemented a fully-connected multi-layer perceptron (MLP) for binary classification of the THA2 dataset.

The network architecture is:

- Input layer: 2 neurons (features X_0, X_1)
- Hidden layer 1: 10 neurons, ReLU activation
- Hidden layer 2: 10 neurons, ReLU activation
- Output layer: 2 neurons, softmax activation (class probabilities)

Used the **cross-entropy** loss between the one-hot labels and the **softmax** outputs.

The data is split into:

- **Training set:** 410 samples
- **Validation set:** 82 samples

Before training, we **standardised** each feature using the training statistics:

$$x' = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

$$x' = \sigma_{\text{train}} \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

The same mean and std are then applied to the validation set.

Hyperparameter choices (Point 2)

We needed to choose a learning rate, batch size, and parameter initialisation.

The MLP is trained with **stochastic gradient descent** (**mini-batch SGD**) using our own backprop implementation.

1. Learning rate (η)

Tested $\eta \in \{0.1, 0.01, 0.001\}$ with full-batch training and initialisation $\sigma = 0.1$.

- $\eta = 0.001$: optimisation progressed very slowly and validation loss stayed close to 0.69.
- $\eta = 0.01$: higher accuracy but still relatively high validation loss.
- $\eta = 0.1$: fastest convergence and significantly lower validation loss.

After standardising the data, $\eta = 0.1$ consistently gave the best combination of convergence speed and final validation loss, so we **selected $\eta = 0.1$** .

2. Batch size (B)

With $\eta = 0.1$ and $\sigma = 0.1$ we compared batch sizes {full batch (410), 16, 4}.

- Full-batch: validation loss ≈ 0.64 , accuracy $\approx 78\%$.
- $B = 4$: similar accuracy but slightly higher loss and noisier training.
- **$B = 16$** : lowest validation loss (≈ 0.18 in the sweep) and accuracy $\geq 95\%$.

We therefore chose **batch size $B = 16$** for the final model.

3. Initialisation scale (σ)

Initialised weights from $N(0, \sigma^2)N(0, \sigma^2)$ and biases to zero.

With $\eta = 0.1$ and $B = 16$ we tested $\sigma \in \{0.01, 0.1, 1.0\}$.

- All three values converged, but
- $\sigma = 0.1$ consistently produced the **lowest validation loss (~ 0.14)** while keeping accuracy very high ($\approx 97.6\%$).

We therefore used **$\sigma = 0.1$** as the standard deviation for weight initialisation.

In summary, the final training configuration is:

- **Architecture:** 2-10-10-2, ReLU + **softmax**
- **Loss:** **cross-entropy**
- **Learning rate:** $\eta = 0.1$
- **Batch size:** $B = 16$ (mini-batch SGD)
- **Initialisation:** $w \sim N(0, 0.12)$ $w \sim N(0, 0.12)$, $b = 0$ $b = 0$
- **Preprocessing:** feature standardisation (per feature, using training mean/std)

Training behaviour and stopping criterion (Point 3)

During training we recorded the loss at each epoch on both the **training** and **validation** sets.

- At the start (epoch 0-5), both losses are around ~ 0.69 (random predictions).

- By about **20 epochs**, the losses drop sharply to ~ 0.18 .
- After that, both curves decrease slowly and **stabilise between 0.13 and 0.15**.
- The gap between training and validation loss remains small (~ 0.03), indicating only mild overfitting.

We trained for **200 epochs**. Around 40–60 epochs the validation loss is already almost flat, and by 150–200 epochs additional improvements are marginal. We therefore consider 200 epochs “sufficient training” and use this point as our final model.

The “Training vs Validation Loss” plot that we will include in the report shows:

- Both curves decreasing monotonically.
- Validation curve closely tracking the training curve, with no sign of divergence.

This satisfies the requirement to *“plot the loss on the training and validation sets and justify the stopping criterion”*.

Final validation performance

With the selected hyperparameters ($\eta = 0.1$, $B = 16$, $\sigma = 0.1$) and standardised inputs, the final model achieves on the **validation set (82 samples)**:

- **Validation loss:** 0.1292
- **Validation accuracy:** 0.9756 (80/82 correct)
- **Confusion matrix (rows = true class, cols = predicted class):**

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This means:

- 40 correctly classified class-0 samples, 1 class-0 $\rightarrow$ class-1 error.
- 40 correctly classified class-1 samples, 1 class-1 $\rightarrow$ class-0 error.

Errors are extremely rare and roughly balanced between the two classes, which indicates a well-behaved classifier on this dataset.